Network Layer

Routing Algorithm

Distance Vector Routing

Lecture 12 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

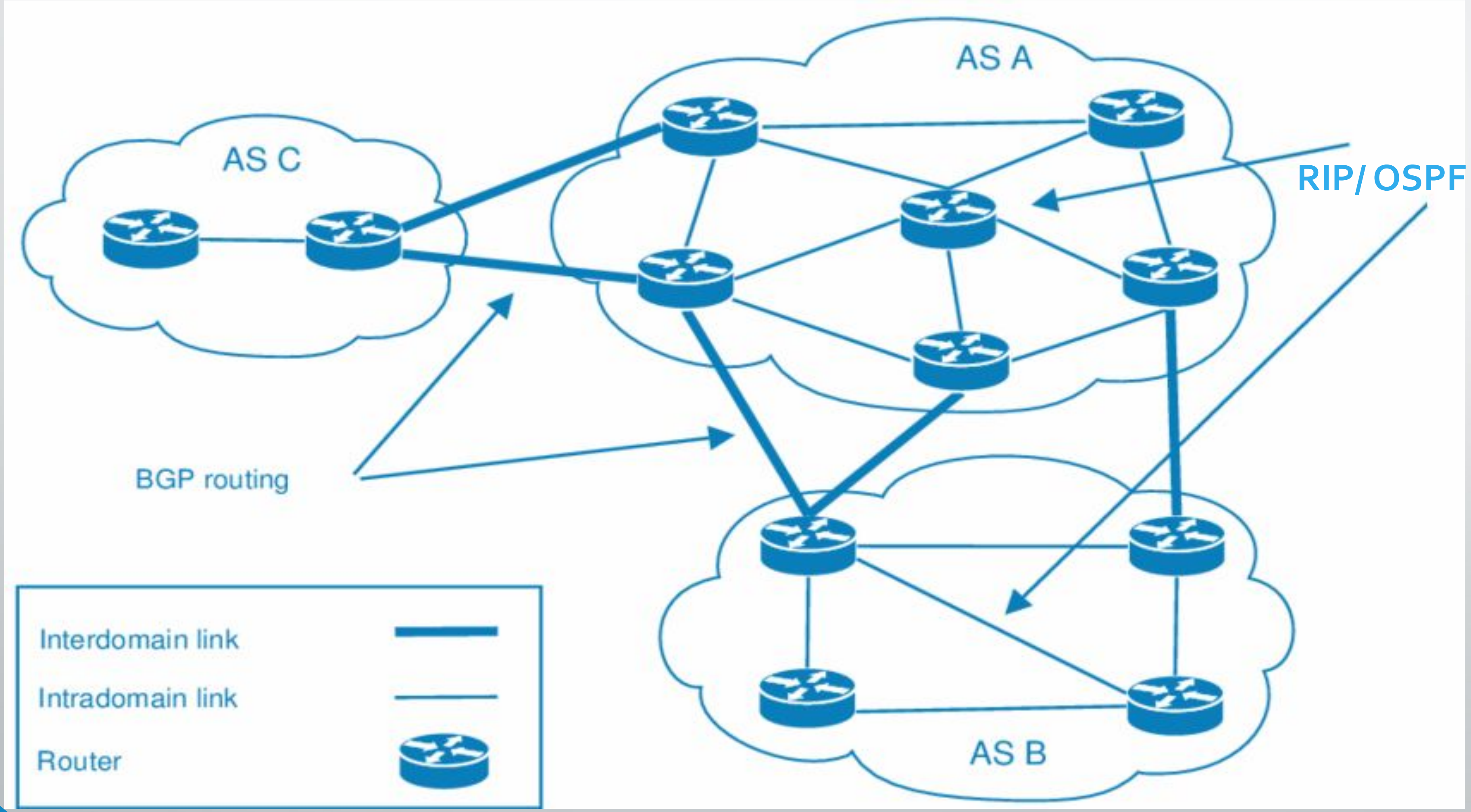
Objectives

- understand principles behind network layer services:
  - routing algorithms
    - **distance vector**
    - **link state**

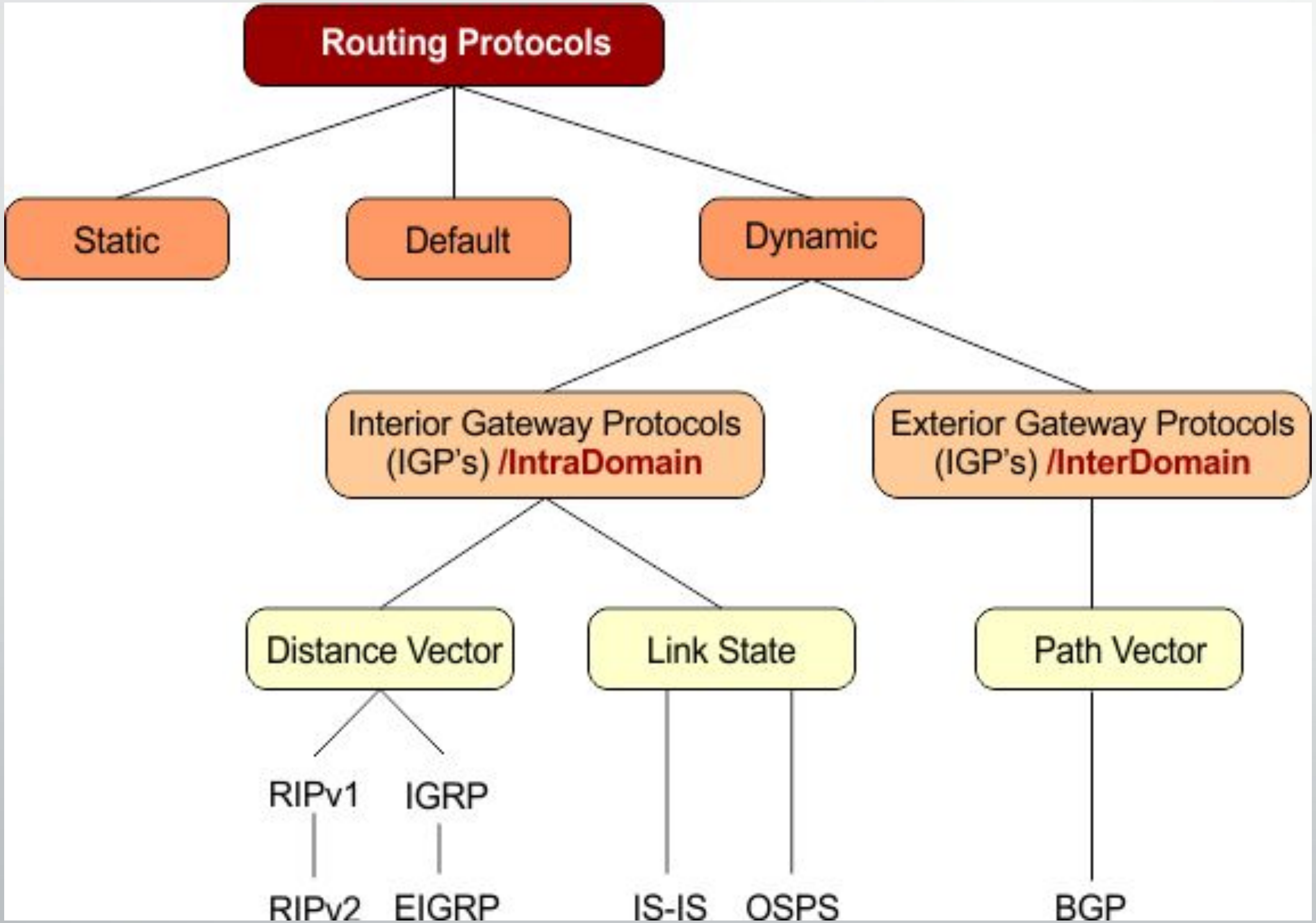
Autonomous Systems

- Internet is divided into autonomous systems (AS).
- An AS = a group of networks and routers controlled by one organization.
- Routing ***inside*** an autonomous systems(AS) is called **intra-domain routing**.
- Routing ***between*** autonomous systems(AS) is called **inter-domain routing.**

Autonomous Systems



Popular Routing Protocols

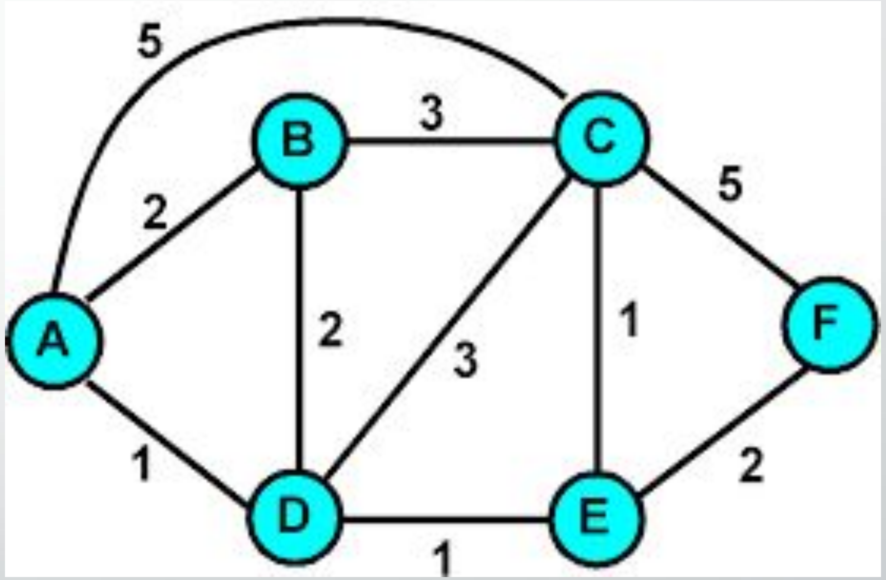


Routing Algorithms

- Given a set of routers and links connecting the routers.

- Routing algorithm finds a "good" path from the source to destination router.

- Good path = Least cost path



Routing Algorithm classification

Global:

- Every router knows the whole network (all nodes, all links, and their costs).
- Each router builds a complete map (topology).
- Uses **link state algorithms** (like Dijkstra's shortest path).

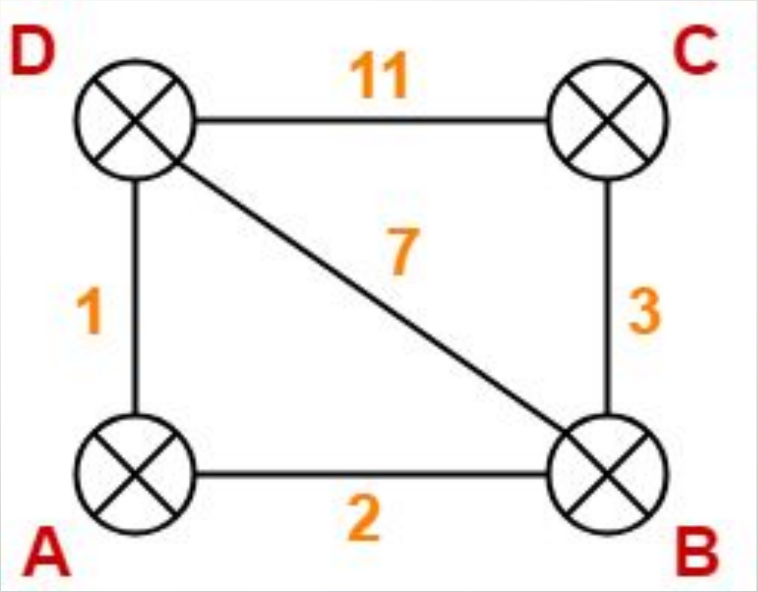
Decentralized:

- Each router only knows about its neighbors and the cost to reach them.
- Routers share information with neighbors and update step by step.
- Uses **distance vector algorithms** (like Bellman-Ford).

Distance Vector Algorithm

Distance vector algorithm

- Based on **Bellman-Ford Algorithm**.
- Each router finds **shortest paths** to all destinations.
- Shortest path cost can be:
  - Hop count
  - Link delay
  - Bandwidth



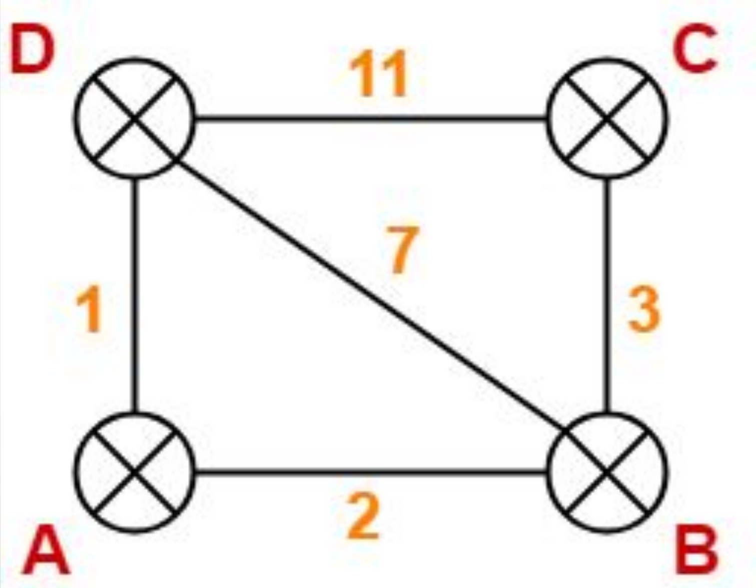
- Uses a **weighted network (topology)**.

How Does It Work?

- Each router keeps a **Distance Vector (table)**.
- Router **shares its table** with neighbors regularly.
- When a router gets a table from a neighbor, it **updates its own table** using the Bellman-Ford rule.
- Repeat until all routers know the shortest paths.

NODE D Table

Node	Distance
A	1
B	7
C	11
D	0



NODE C Table

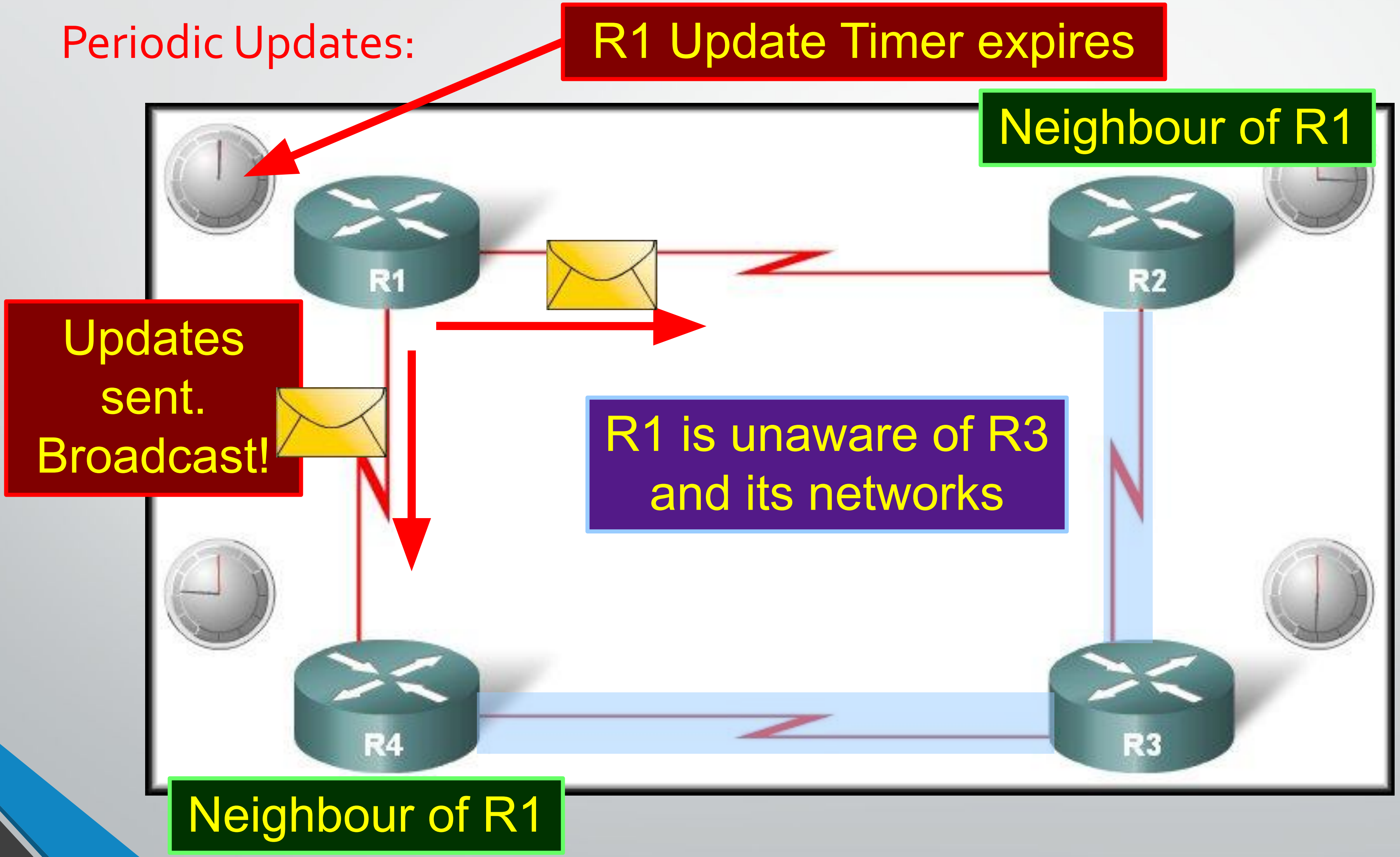
Node	Distance
A	No Info
B	3
C	0
D	11

Periodic Updates of Distance Vector

- Routers regularly send their **entire routing table** to neighbors (e.g., every 30 seconds in RIP).
- This is **simple but inefficient**.
- Each Router is only aware of the:
  - The addresses of its **own interfaces**.
  - The addresses of **neighbors** running the

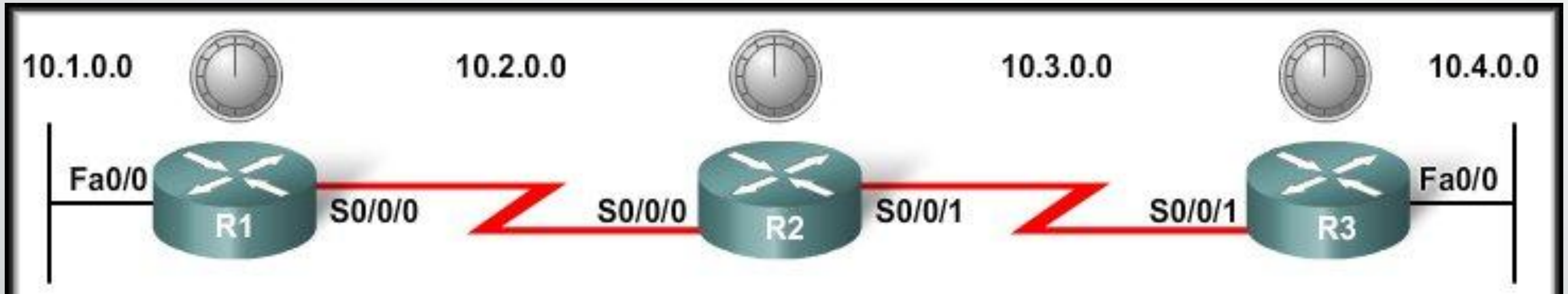
Operation of Distance Vector

Periodic Updates:



Distance Vector Routing Protocols

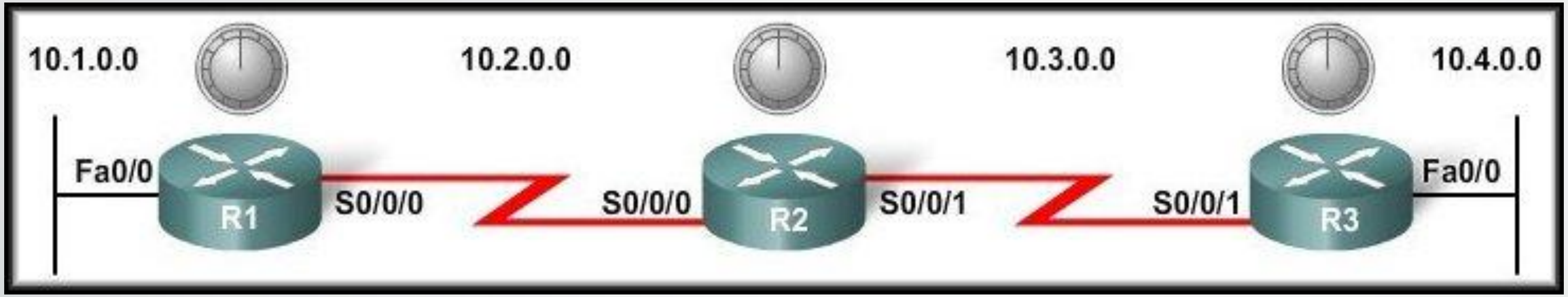
Network Discovery



Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

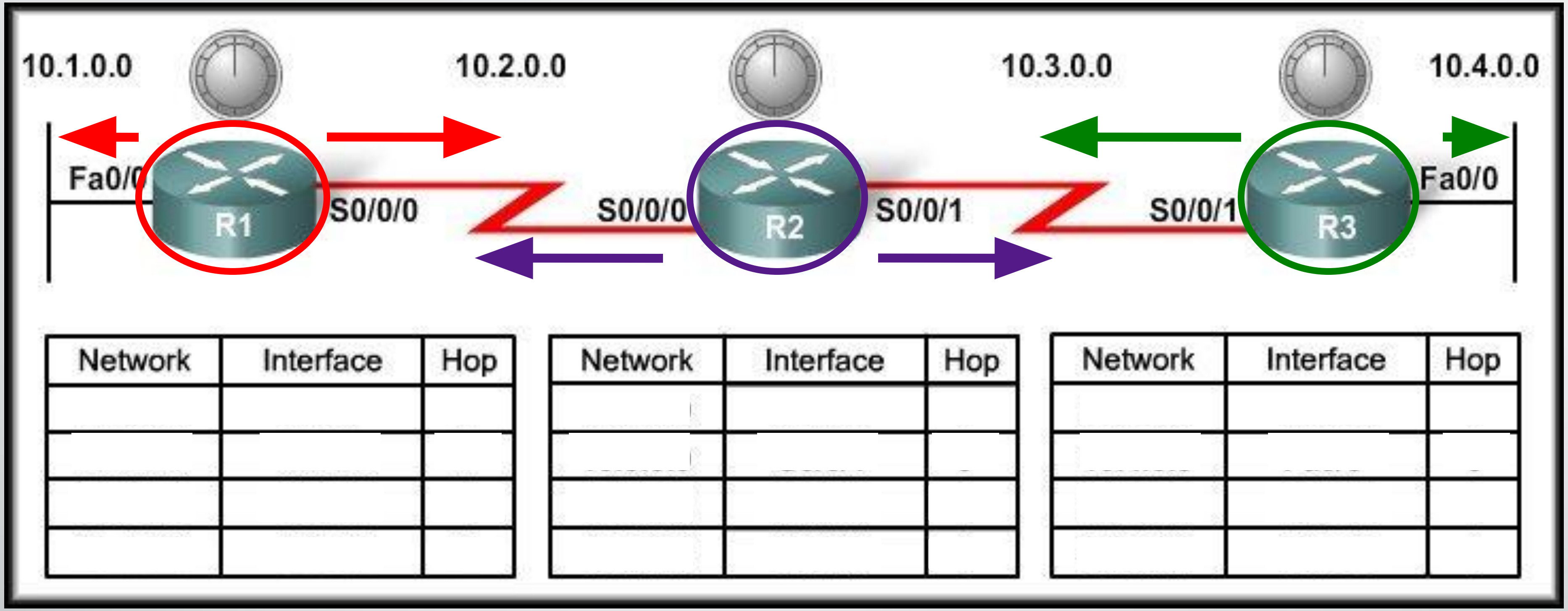
Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2



- **Network Discovery:**
  - Is part of the process of the routing protocol algorithm that enables routers to learn about remote networks for the first time.

Cold Start



- **When a router powers up:**
  - Knows nothing about the network topology.
  - Knows only the information saved in NVRAM.
  - Sends updates about its known networks out all ports.

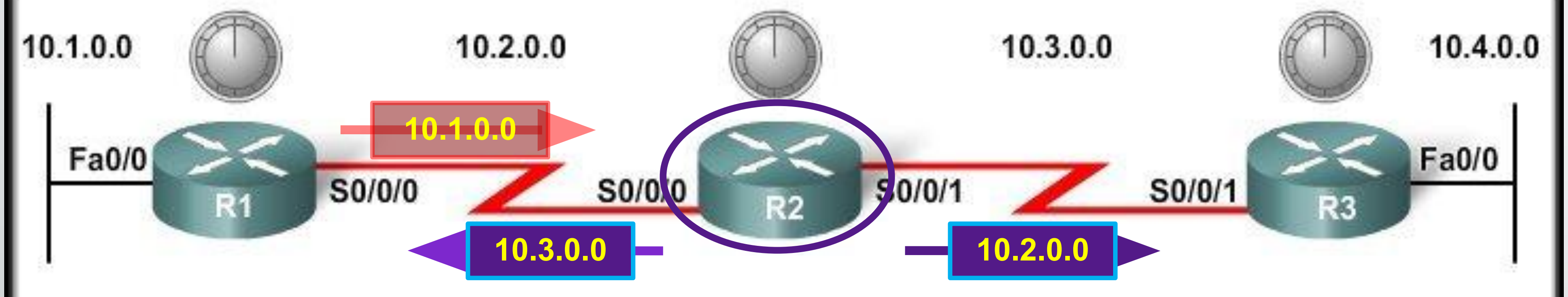
Initial Exchange of Routing Information

Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0

Initial Exchange of Routing Information

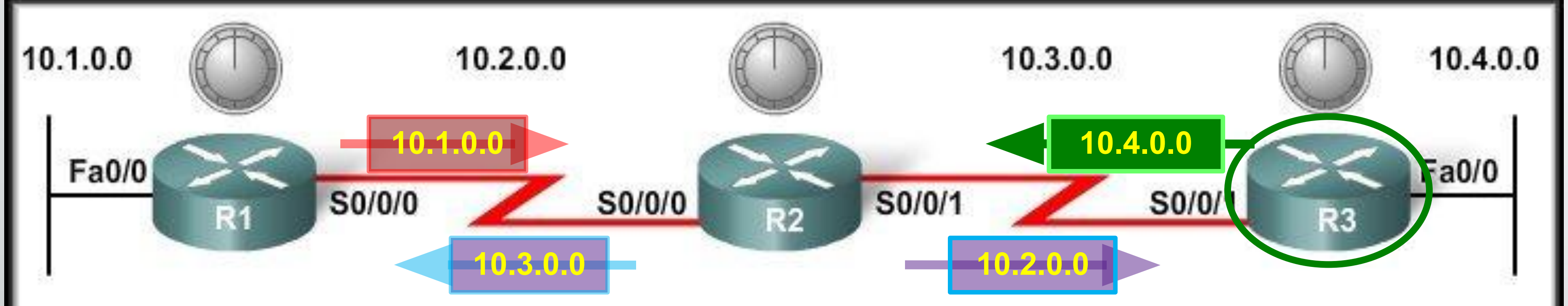


Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0

Initial Exchange of Routing Information

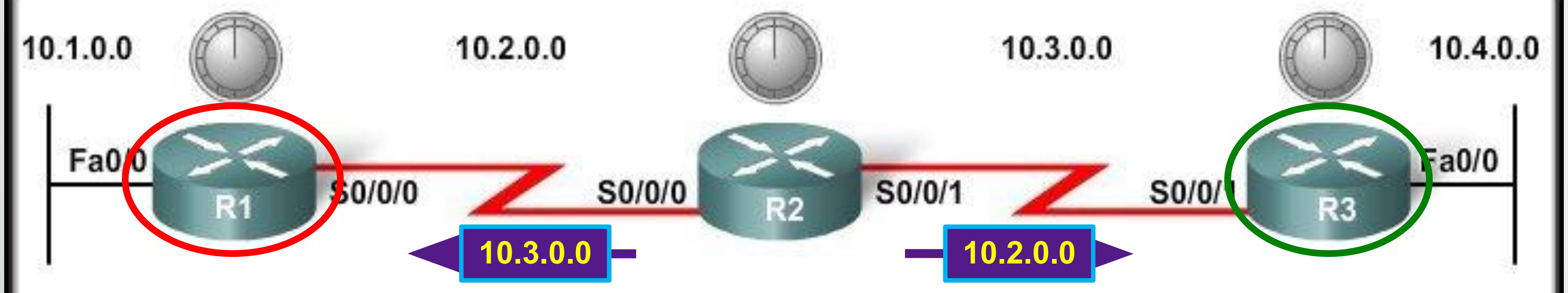


Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0

Initial Exchange of Routing Information

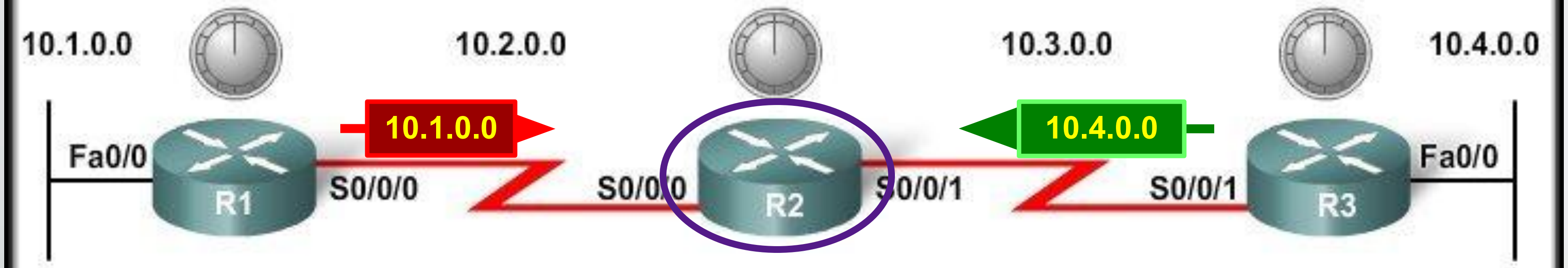


Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0

Initial Exchange of Routing Information



Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1

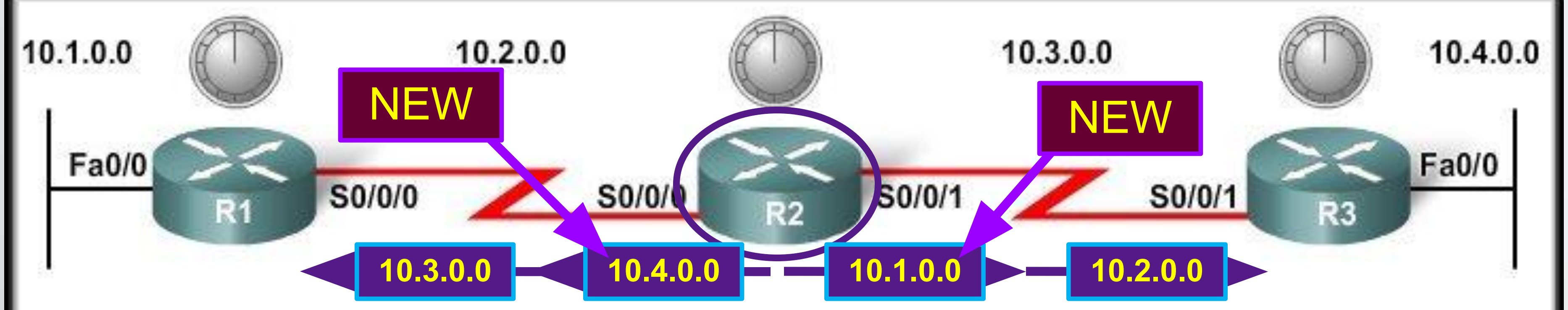
Next Exchange of Routing Information

Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1

Next Exchange of Routing Information



Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1

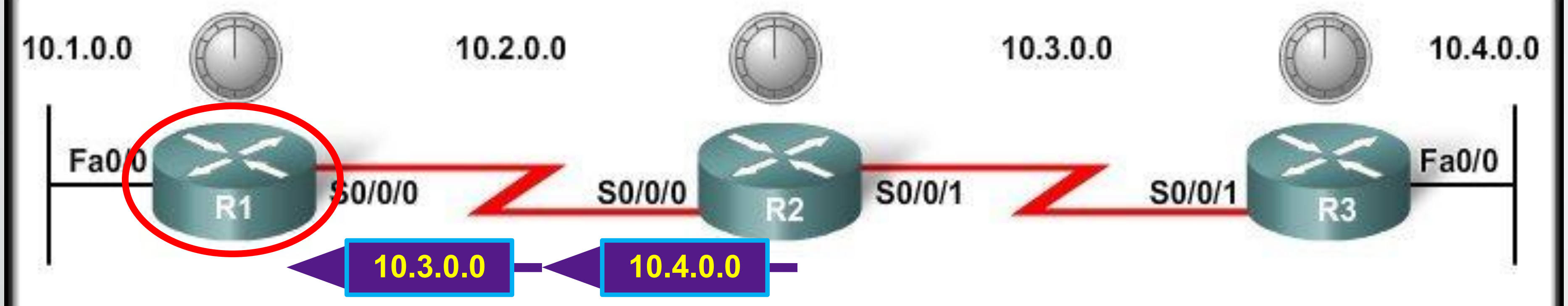
Next Exchange of Routing Information

Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1

Next Exchange of Routing Information

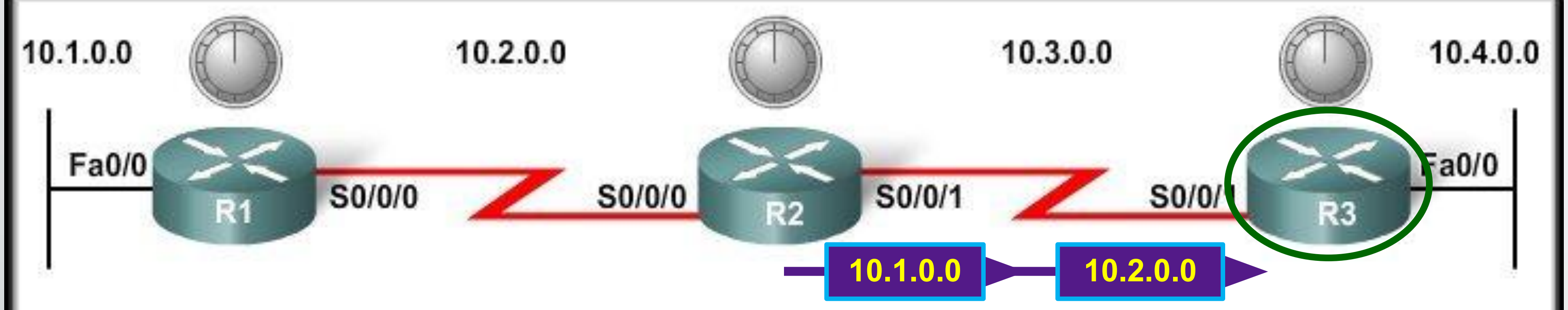


Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1

Next Exchange of Routing Information



Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1

Next Exchange of Routing Information

Network	Interface	Hop
10.1.0.0	Fa0/0	0
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/0	1
10.4.0.0	S0/0/0	2

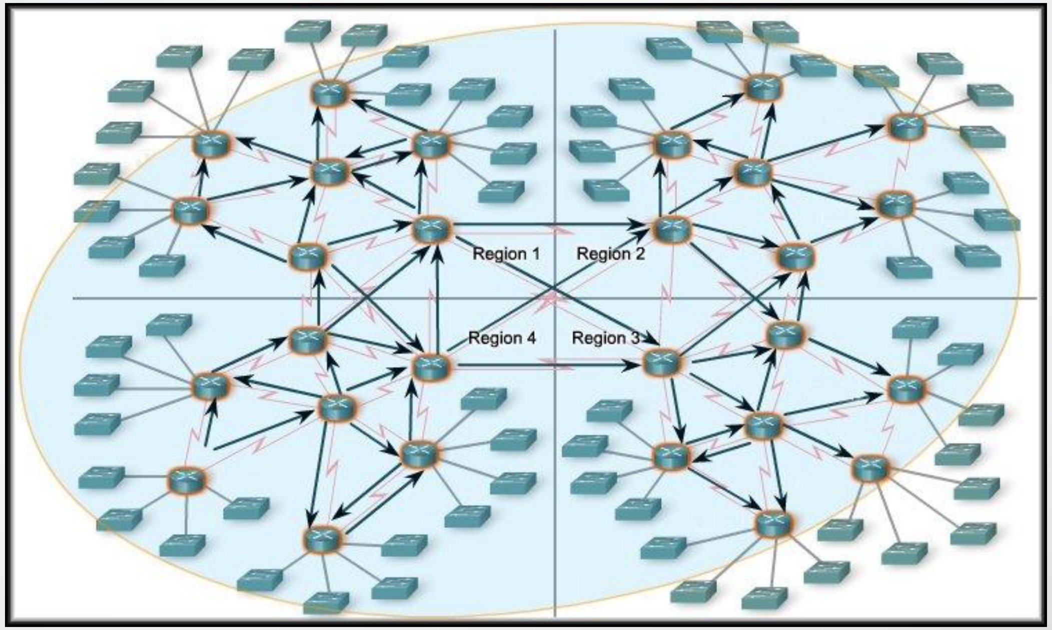
Network	Interface	Hop
10.2.0.0	S0/0/0	0
10.3.0.0	S0/0/1	0
10.1.0.0	S0/0/0	1
10.4.0.0	S0/0/1	1

Network	Interface	Hop
10.3.0.0	S0/0/1	0
10.4.0.0	Fa0/0	0
10.2.0.0	S0/0/1	1
10.1.0.0	S0/0/1	2

- The network has **CONVERGED!**
  - All routers now know about all of the networks attached to all of their neighbouring routers.

Convergence

- The amount of time it takes for a network to converge is directly proportional to the size of that network.



- Routing protocols are compared based on how fast they can propagate this information - their speed to convergence.

- A network is not completely operable until it has converged.
    - Network administrators prefer routing protocols with shorter convergence times.

THE END